



Ontogenetic and Geographic Phytochemical Variation in *Mimulus moschatus*, a Perennial Monkeyflower

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Abstract

Secondary metabolites are critical to plant defenses and frequently exhibit variation among populations associated with heterogeneous climatic factors and/or herbivore communities. We examined genetic-based geographic and ontogenetic variation in phenylpropanoid glycoside (PPG) concentrations and arsenals in *Mimulus moschatus*, a perennial monkeyflower native to western North America. We conducted a greenhouse common garden with maternal lines collected from three populations sampled along a latitudinal gradient paralleling spatial sampling from a previous study in the closely related congener, *Mimulus guttatus*. We assessed PPG arsenals as well as concentrations of individual and total PPGs within leaf tissue from two developmental stages. Populations differed in total concentration and arsenal composition of PPGs, with the populations from intermediate latitudes exhibiting higher concentrations than higher and lower latitude populations. Leaves from earlier developmental stages had higher overall PPG concentrations than leaves from later developmental stages across all populations, suggesting juvenile tissue is better defended than more mature tissue. Spatial patterns in relative PPG concentrations and arsenal composition tightly paralleled those in the closely related *M. guttatus* across space despite consistently lower total PPG concentrations in *M. moschatus*. Because we did not measure herbivory and our sample size is limited, these results should be interpreted cautiously. Nonetheless, they provide baseline data for understanding geographic and developmental variation in chemical defenses and generate hypotheses about environmental influences and potential convergence in defenses among closely related species.

Keywords Plant chemical defense · Phenylpropanoid glycosides · *Erythranthe moschata* · Natural selection · Geographic variation · Convergent evolution

Introduction

Secondary metabolites are essential components of plant defenses, providing protection against herbivores, pathogens, and environmental stressors (Hanley et al. 2007; Rasemann and Agrawal 2009; Erb et al. 2012). These compounds are found in all higher plants (Fraenkel 1959; Wink 2003) and often vary significantly across populations through differences in local herbivore communities and environmental conditions (Woods et al. 2012; Moore et al. 2014; Anderson

et al. 2015; Abdala-Roberts et al. 2016). Phytochemical defenses can vary in total abundances of specific classes of compounds (Feeny 1976; Rasemann and Agrawal 2011; Coley et al. 2019; Wetzel and Whitehead 2020), in relative abundances of different classes of compounds (Rhoades and Cates 1976; Richards et al. 2015; Anstett et al. 2016a; Blanchard and Bowers 2020) or relative abundances of different compounds within a single class of chemical defenses (Berenbaum 1983; Glassmire et al. 2016; Blanchard and Bowers 2020; Holeski et al. 2021). Novel combinations and/or variation in relative abundances of phytochemical compounds among populations are referred to as defense arsenals.

Different plant defense hypotheses have been used to explain spatial and temporal variation in defense compounds. The optimal defense theory (ODT; Rhoades 1979) suggests that plants allocate defensive resources to tissues based on their value and vulnerability, prioritizing organs

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with higher fitness importance, such as reproductive structures and juvenile leaves (McKey 1974; Zangerl and Bazzaz 1992). Among populations, trade-offs between growth, reproduction, and defense may evolve in response to variation in environmental quality and resources (Stamp 2003). For example, in *Mimulus* species, populations experiencing longer growing seasons tend to invest more resources into constitutive chemical defense traits relative to populations with shorter growing seasons (Holeski et al. 2013, 2021; Kooyers et al. 2017, 2020). The latitudinal herbivory-defense hypothesis (LHDH; Coley and Aide 1991), on the other hand, posits that herbivory pressure increases towards the equator, leading to the evolution of greater investment in defensive traits by tropical plants compared to their temperate counterparts (Coley and Aide 1991; Schemske et al. 2009). As a result, more tropical populations are predicted to have higher concentrations and a greater variety of secondary metabolites (Ehrlich and Raven 1964; Moles et al. 2011; Marquis et al. 2016).

While there is spatial variation among populations in arsenals and abundances of defenses, ontogenetic variation, or changes in defense allocation as plants progress through development, also plays a crucial role in plant defense. Plants tend to allocate more resources to defense at juvenile developmental stages, protecting vulnerable tissues from herbivory, while more mature plants shift their resources toward reproduction, reducing investment in defense traits (Zangerl and Bazzaz 1992; Boege and Marquis 2005; Quintero and Bowers 2011; Barton and Boege 2017). Temporal shifts in resource allocation align with the ODT, which predicts that defense allocation will change over time based on the cost of defending different tissues (McKey 1974; Rhoades 1979; Stamp 2003). However, differences in reproductive strategies play an important role in shaping ontogenetic defense patterns (Boege and Marquis 2005), thus, it remains difficult to generalize these patterns across plant species (Kessler and Halitschke 2009; Barton and Koricheva 2010; McCall and Fordyce 2010).

Nearby populations are expected to share similar defense arsenals if they experience comparable biotic and abiotic environments (Agrawal et al. 2006; Abdala-Roberts et al. 2016; Jameel et al. 2025). Variation in herbivore communities, climate and resource availability can drive divergence in plant defenses (Anstett et al. 2015; Hahn and Maron 2016; Moreira et al. 2018; López-Goldar et al. 2020; Rotter et al. 2022). Climate can also influence herbivore species richness and abundance, as well as shape plant growth and life-history traits (Kambach et al. 2016; Bont et al. 2020). Thus, populations in close proximity might evolve similar defense arsenals due to shared selection pressures imposed by herbivore communities (Mauricio and Rausher 1997; Richards et al. 2015; Salazar et al. 2018), reflecting broader

patterns of spatial variation in plant defense traits (Anstett et al. 2015; Kooyers et al. 2017; Rotter et al. 2022). Similarly, classic theory proposes that similar selection pressures from herbivores should lead to convergence in phytochemical defenses among closely related species (Ehrlich and Raven 1964; Barrett 1987; Agrawal and Fishbein 2006). However, heterogeneity in herbivore pressure or communities may drive the evolution of divergent chemical arsenals, even over short geographic scales (Berenbaum 1978, 1983; Coley et al. 2018). For instance, divergence in patterns of chemical defenses has been documented in tropical systems like *Inga*, where species develop unique phytochemical profiles despite being closely related (Coley and Kursar 2014). These patterns suggest that both phylogenetic constraints and ecological pressures interact to shape defense strategies, with the potential to drive chemical divergence and even, potentially, speciation (Endara et al. 2015; Marquis et al. 2016).

Utilising *Mimulus moschatus*, a broadly distributed perennial forb, we use a common garden experiment to examine genetic-based patterns of variation in concentrations of ecologically important secondary metabolites across ontogeny and among populations. Specifically, we ask: (1) how do total concentrations and arsenals of defense compounds vary geographically? (2) how does ontogeny influence both total concentrations and the composition of defense compounds? (3) do neighbouring populations of congeneric species share similar defense arsenals and concentrations of defense compounds? Our study seeks to better understand how geography, ontogeny and evolutionary history shape genetic-based patterns of plant defense traits within and among populations. We emphasize that our design provides a first, conservative evaluation of patterns since we surveyed limited populations and did not assess herbivory. Accordingly, our interpretations are offered as hypothesis-generating rather than definitive tests of classic hypotheses.

Materials and Methods

Study System

Mimulus moschatus (syn: *Erythranthe moschata* [Dougl. ex Lindl.] G.L.Nesom [Phrymaceae]) is a perennial herb native to the western United States and Canada (Grant 1924; Whittall et al. 2006). It typically inhabits moist, shaded to unshaded environments such as stream banks, seeps, and wet meadows, often at mid-to-high elevations. The species is characterized by its creeping growth habit, glandular-pubescent foliage, and yellow, bilaterally symmetrical flowers. It reproduces both clonally via rhizomes

and sexually through insect-mediated pollination, with bees being common visitors. It shares a similar widespread distribution with its well-studied model congener, *M. guttatus* (syn: *Erythranthe guttata* [Fisch. DC.] G.L.Nesom [Phrymaceae]; Vickery 1995; Beardsley et al. 2004; Wu et al. 2008; Barker et al. 2012; Lowry et al. 2019), but occurs in habitats that often begin drier, but dry out slower than *M. guttatus* habitats.

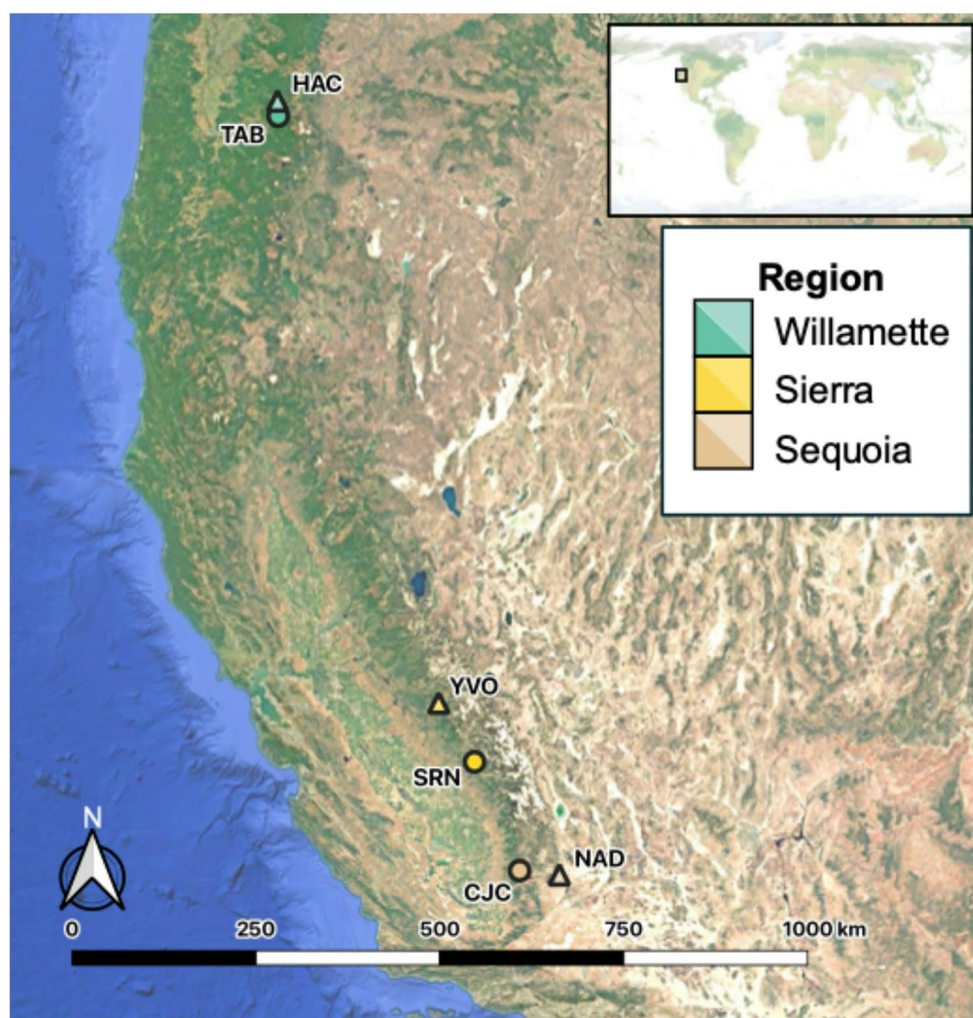
Phenylpropanoid glycosides (PPGs; phenylethanoid glycosides, Jiménez and Riguera 1994; or caffeic acid esters, Mølgaard and Ravn 1988) are ecologically important chemical defense compounds that are most abundant among species of the Lamiales order, but also present across dicotyledons (Mølgaard and Ravn 1988; Jiménez and Riguera 1994; Holeski et al. 2021). Biosynthesis of these defense compounds begins in the shikimic acid-phenylpropanoid pathway and the resulting hydroxycinnamic acids are bound to sugar moieties by ester bonds and to hydroxyphenylethyl moieties via glycosidic linkages (Herrmann and Weaver 1999; Vogt 2010; Keefover-Ring et al. 2014). The geographic distribution and evolutionary ecology of these and

other PPGs have been best studied in *M. guttatus*, but have also been studied throughout the genus (Holeski et al. 2021). Populations of *M. guttatus* vary qualitatively and quantitatively in both constitutive and induced concentrations of their PPGs (Holeski et al. 2013; Keefover-Ring et al. 2014; Kooyers et al. 2017). Studies in *M. guttatus* also suggest that most PPGs deter generalist lepidopterans and molluscs while some may act as feeding stimulants for specialist lepidopterans (Holeski et al. 2013; Rotter et al. 2018). In other species, these same PPGs function as deterrents against generalist herbivores (Mølgaard 1986). The presence and arsenals of PPGs in *M. moschatus* have not yet been described.

Sampling and Extraction

Seeds from *M. moschatus* were collected from three populations spanning ~1000 km of latitude across the Sierra Nevada and Cascades (Fig. 1) during the summer of 2021. Eight individuals from each population (total individuals $n=24$) were grown under greenhouse conditions at the University of Louisiana at Lafayette for approximately 100 days.

Fig. 1 Relief map depicting locations of sampled *Mimulus* populations across the western United States. Eight maternal lines were taken from each *M. moschatus* population (circles; TAB, SRN and NAD; total $n=24$). *Mimulus guttatus* populations from Kooyers et al. (2017) are in lighter shades than nearby *M. moschatus* populations (triangles; HAC: $n=6$, YVO: $n=9$, NAD: $n=8$ maternal lines). The inset shows the relative location of focal map. Population names are also indicated on the map next to each point



Seeds were germinated in 142 mL pots (Hummert International, Erath City, MO, USA) until the second pair of true leaves unfurled then transplanted to 475 mL pots (Hummert International, Erath City, MO, USA) filled with Metro-Mix 830 (Sun Gro Horticulture, Agawam, MA, USA). Plants were randomized within the greenhouse in 10" × 20" trays (Greenhouse Megastore, Danville, IL, USA) under 18-hour day: 6-hour night light cycles. Leaf samples were simultaneously taken from the earliest (3rd to 5th true leaf pair) and the latest (8th to 10th true leaf pair) fully expanded leaf pairs of each plant (samples $n=48$). The earliest and latest fully expanded leaves are hereafter referred to as juvenile (early) and mature (late) tissue, respectively.

Concentrations of PPGs within leaf tissue samples were assessed using high-performance liquid chromatography (HPLC). Briefly, plant tissue was freeze-dried at $-50\text{ }^{\circ}\text{C}$ and 0.039 mbar in a FreeZone 4.5–50 $^{\circ}\text{C}$ freeze direr (Lab-conco, Kansas City, MO, USA), ground, and weighed prior to shipping to Northern Arizona University. There, PPGs were extracted as described previously (Holeski et al. 2013, 2014) and quantified via HPLC (Agilent 1260 HPLC) with a diode array detector and Poroshell 120 EC-C18 analytical column ([4.6 × 250 mm, 2.7 μm particle size]; Agilent Technologies, Santa Clara, CA, USA) maintained at 30 $^{\circ}\text{C}$. HPLC run conditions included a binary mobile phase gradient with 3.1 mmol/L phosphoric acid as mobile phase A, and acetonitrile as mobile phase B at a constant total flow rate of 0.75 mL/min. The gradient for each run consisted of B initially set at 5%, 15% at 5.5 min, 16% at 7.5 min, 13% at 9 min, 20% from 10 to 12 min, 60% at 13 min, and 95% from 14 to 15 min. From 15 to 16.5 min, mobile B returned to 0% to re-equilibrate the column. Three mL of the standards and samples were injected, monitored for ultraviolet signals at 274 (catechol) and 340 nm (PPGs), and a diode array detector was used to collect ultraviolet data from 190 to 400 nm. Quantitative estimates of PPGs were calculated as verbascoside equivalents, using a standard solution of pure verbascoside (Santa Cruz Biotechnology, Dallas, TX, USA) as previously described (Holeski et al. 2013, 2014). Total PPG concentrations (hereafter ‘total PPGs’) were calculated as the sum of all individual PPG concentrations.

Statistical Analyses

All statistical analyses, data manipulation and visualization were performed in R v. 4.4.1 (R Core Team 2024). Data manipulation was facilitated with *tidyverse* v. 2.0.0 packages (Whickham et al. 2019) and model results were visualized with *ggplot2* v. 3.5.1 (Wickham 2016).

Geographic and Ontogenetic Variation in Defenses

To assess geographic and ontogenetic variation in concentrations and arsenals of phytochemical defenses in *M. moschatus*, we used a series of univariate linear models testing the effects of population and ontogeny. First, we fit concentrations of each individual PPG and total PPGs to models with “population”, “ontogeny” and their interaction with the *lm* function and assessed statistical significance with a type-III ANOVA using the *Anova* function in the *car* package v. 3.1.2 (Fox and Weisberg 2019). Individual PPG concentrations were $\log(x+1)$ transformed to improve model fit. Post-hoc Tukey tests were used to identify the direction and magnitude of changes in our models with the *emmeans* function in the *emmeans* package v. 1.10.2 (Lenth 2024).

To examine the composition of PPG arsenals, we performed pairwise Pearson correlations between individual and total PPGs with the *cor* function and assessed statistical significance with *cor.test*. We examined how PPG arsenals differed across ontogeny and between populations by fitting a dissimilarity matrix, generated with the *vegdist* function in the *vegan* package v. 2.6–10 (Oksanen et al. 2025), to a model with “population”, “ontogeny” and their interaction. Statistical significance was assessed with a PERMANOVA using the *adonis2* function in the *vegan* package. Additionally, we created univariate linear models, fit with the *lm* function, with the two first NMDS axes from our dissimilarity matrix as response variables to further dissect how PPG arsenals changed across populations and ontogeny. We assessed statistical significance as above.

In our models, “Population” tested how the means of our response variables are predicted to vary across space, and “Ontogeny” (juvenile vs. mature) examined differences in the means of response variables through development. The interaction of “Population × Ontogeny” tested how ontogenetic shifts in our response variables depended on the sampled population.

Congeneric Comparison of Defenses

To assess variation in PPG concentrations and arsenals between neighbouring populations of *M. moschatus* and *M. guttatus* (Kooyers et al. 2017), we leveraged published *M. guttatus* data from Kooyers et al. (2017) which also used a common garden greenhouse experiment to generate their phytochemical data. We used univariate linear models testing the effects of species and region on PPG concentrations and arsenals. First, we selected nearby elevation-matched populations from both species (Fig. 1). This was done by calculating Haversine distances between all populations with the *distm* function from the *geosphere* package v. 1.5.20 (Hijmans 2024) and pairing populations from both

species with similar elevations. Selected population pairs were less than 100 km apart and within 300 m elevation of one another. We fit univariate models with total PPGs and relative concentrations of the two most abundant individual PPGs—conandroside and calceolarioside A for all *M. moschatus* and *M. guttatus* populations—as response variables using the *lm* function. Predictor variables were “Species”, “Region”, and their interaction. Only juvenile leaf concentrations from this study and constitutive concentrations from Kooyers et al. (2017) were used to balance the comparisons. Furthermore, replicates from the same maternal line were averaged from Kooyers et al. (2017) since the current study only had single replicates from each maternal line. We assessed statistical significance as described above. A caveat to note is that a single pair from the 3rd to 5th leaf pairs were used for PPG analyses from the current study whereas all leaves from the 3rd to 5th leaf pairs were used for analyses in Kooyers et al. (2017).

In these models, “Species” tested how the means of our response variables differ between species, and “Region” assessed differences in the means of response variables across space. The interaction of “Species × Region” tested whether observed differences between species varied spatially. We expected total PPG concentrations and relative concentrations of individual PPGs to vary between species if *M. moschatus* and *M. guttatus* differentially allocate resources to defenses or experience selection pressures from divergent herbivore communities. In contrast, we expected no effect of species on congeneric pairs if our paired populations shifted their PPG concentrations and arsenals in

similar ways across our sampled range. Likewise, if abiotic and biotic environmental factors affect defenses, we expected PPG concentrations to differ in similar ways for both species across space.

Results

Mimulus moschatus has many of the same PPGs as *M. guttatus* (Holeski et al. 2013; Keefover-Ring et al. 2014) and other *Mimulus* species (Holeski et al. 2021); however, it also possesses novel PPGs which are unique to this species. Calceolarioside A, calceolarioside B, conandroside, mimuloside and verbascoside have previously been identified in *M. guttatus* (Holeski et al. 2013) as well as other *Mimulus* species (Holeski et al. 2021) and have been structurally characterized (Keefover-Ring et al. 2014). Unknown PPG 16 (Unkn 16) has previously been putatively identified in *M. guttatus*, but has not yet been structurally characterized (Keefover-Ring et al. 2014). Phenylpropanoid glycosides b through h are unknown PPGs which have not yet been structurally characterized nor previously identified in other *Mimulus* species (Table S1). Among *M. moschatus* populations, conandroside and calceolarioside A were the most abundant PPGs while concentrations of other PPGs were much lower, aside from PPG h which was relatively elevated in the southern and middle latitude populations (SRN, CJC; Table S1). Concentrations among populations of the three most abundant PPGs, calceolarioside A, conandroside and verbascoside, were all positively correlated with total PPGs but not with each other (Table S2). All PPGs were present in only the southern population (CJC) while three PPGs were absent from both the middle (SRN; PPGs d, e and f) and northern population (TAB; PPGs b, c and f; Table S1).

Geographic Variation in Defenses of *Mimulus moschatus*

Populations differed in both their total PPGs and PPG arsenals. The middle latitude population (SRN) had higher total PPGs relative to the northern (TAB) and southern (CJC) populations (Population_{Total PPGs}: $F_{2,42} = 27.16$; $P < 0.001$; Table 1; Fig. 2). Specifically, SRN had higher total concentrations than TAB ($\Delta \pm \text{SE}$: 21.78 ± 3.71 mg/g; $t_{42} = 5.870$, $P < 0.001$) and CJC ($\Delta \pm \text{SE}$: 25.22 ± 3.71 mg/g; $t_{42} = 6.795$, $P < 0.001$), while TAB and CJC had similar concentrations ($\Delta \pm \text{SE}$: 3.43 ± 3.71 mg/g; $t_{42} = 0.925$, $P = 0.628$; Fig. 2). Populations also differed in their PPG arsenals (Population_{PPG arsenals}: pseudo $F_{2,42} = 28.00$; $P = 0.001$; Fig. 3), with the middle (SRN) and northern (TAB) populations diverging along NMDS2 and the southern population (CJC) diverging from the other populations along NMDS1. Differentiation

Table 1 Patterns of variation in PPG arsenals and concentrations in *Mimulus moschatus* across ontogeny and populations. Linear models were used for all response variables. ‘Population’ refers to the population from which samples originated. ‘Ontogeny’ contrasted the developmental stages of sampled leaves (juvenile vs. mature). The columns report the degrees of freedom (df), the sums of squares (SS), the *F*-statistic (*F*), and the *P*-value (*P*). Statistically significant results at $P < 0.05$ are highlighted in bold

Response	Factor	df	SS	<i>F</i>	<i>P</i>
Total PPGs (mg/g)	Ontogeny	1	1194.48	10.84	0.002
	Population	2	5984.67	27.16	<0.001
	Ontogeny × Population	2	1.85	0.01	0.992
	Error	42			
NMDS 1	Ontogeny	1	0.58	13.11	<0.001
	Population	2	2.32	26.06	<0.001
	Ontogeny × Population	2	0.10	1.10	0.341
	Error	42			
NMDS 2	Ontogeny	1	0.01	0.57	0.453
	Population	2	2.35	48.98	<0.001
	Ontogeny × Population	2	0.01	0.12	0.889
	Error	42			

Fig. 2 Differences in total phenylpropanoid glycoside levels across populations and ontogeny. Box colours represent populations and box outlines and point shapes represent ontogeny levels. Upper and lower hinges of boxes correspond to the interquartile ranges (IQR), the line inside boxes to medians and the whiskers extend to the most extreme points within $1.5 \times \text{IQR}$ from box hinges

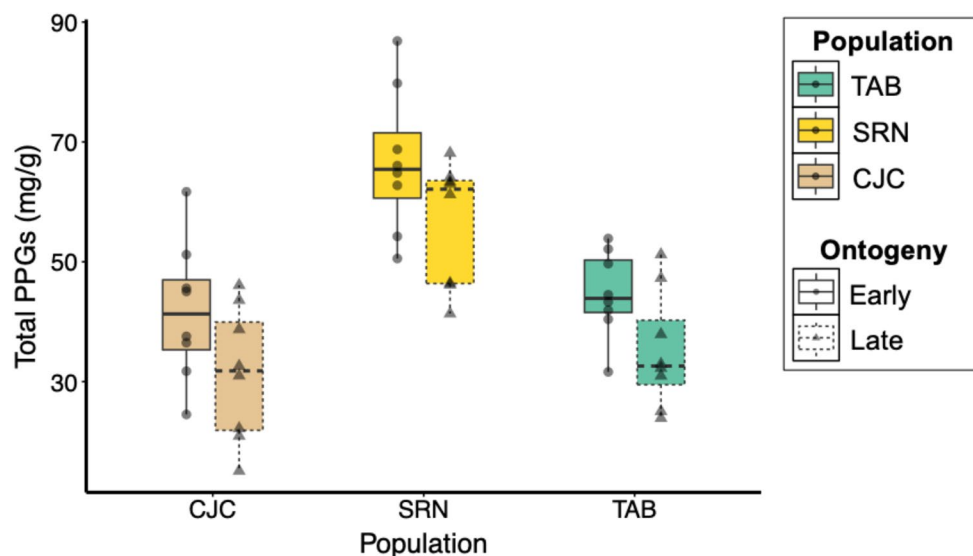
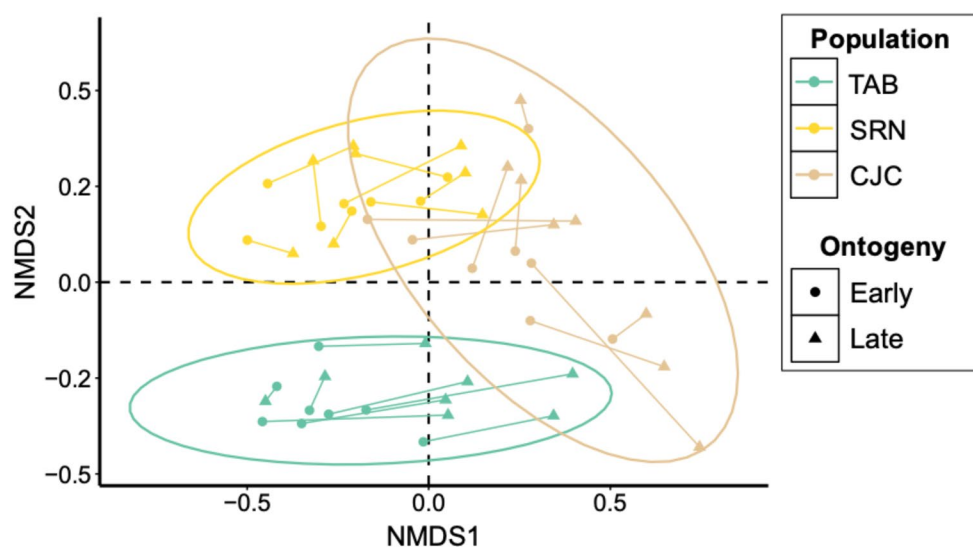


Fig. 3 Composition of PPG arsenals across populations and ontogeny. Samples of different developmental stages are linked by a straight line for each individual with shapes designating leaf developmental stage. Colours represent populations. Larger circles around points represent the 95% confidence ellipses



along NMDS1 was driven by the middle (SRN) and northern (TAB) latitude populations having more conandroside than the southern (CJC) population (Table S1), which loads negatively on this axis (Figure S1; Table S3). The southern (CJC) population also had more PPGs that were typically found at low concentrations (PPG b, c, d and f; Table S1), which load positively on NMDS1 (Figure S1; Table S3). The higher positions of the middle (SRN) and southern (CJC) populations along NMDS2 were mainly driven by their higher concentrations of calceolarioside A relative to the northern (TAB) population, which loads positively on NMDS2 (Figure S1; Table S3). The differentiation of the northern latitude (TAB) population on NMDS1 and NMDS2 was driven by this population having more calceolarioside B than the other two populations (Table S1), which loads negatively on both NMDS axes (Figure S1; Table S3). Within populations, the southern population (CJC) varied

along NMDS2, whereas both the middle and northern populations (SRN and TAB) varied along NMDS1.

Ontogenetic Variation in Defenses of *Mimulus moschatus*

Ontogeny influenced both total PPGs and PPG arsenals among populations. On average, individuals produced 19.55% ($\Delta \pm \text{SE} = 9.98 \pm 3.03 \text{ mg/g}$; $t_{42} = 3.29$; $P = 0.002$) more PPGs per gram of dry leaf weight early in development compared to later in development (Ontogeny_{Total PPGs}: $F_{1,42} = 10.84$; $P = 0.002$; Table 1; Fig. 2). This shift in total PPGs was consistent across populations (Ontogeny $\times$ Population_{Total PPGs}: $F_{2,42} = 0.001$; $P = 0.992$; Table 1; Fig. 2). Similarly, ontogeny also influenced composition of PPG arsenals (Ontogeny_{PPG arsenals}: pseudo $F_{1,42} = 15.04$; $P = 0.001$; Fig. 3), and ontogenetic shifts in PPG arsenals were similar among

Table 2 Model results examining variance in total PPG and key PPG concentrations across species and space. Linear models were used for all response variables. ‘Region’ refers to the part of the sampled range from which congeneric populations originate (Willamette, Sierra and Sequoia). The ‘Species’ factor corresponds to the species used in the comparisons (*M. moschatus* vs. *M. guttatus*). Relative calc A and relative Conand refers to the relative concentration of calceolarioside A (mg/g) and Conandroside (mg/g), respectively. The columns report the degrees of freedom (df), the sums of squares (SS), the *F*-statistic (*F*), and the *P*-value (*P*). Statistically significant results at $P < 0.05$ are highlighted in bold

Response	Factor	df	SS	<i>F</i>	<i>P</i>
Total PPGs (mg/g)	Region	2	8482.35	13.78	<0.001
	Species	1	14264.02	46.34	<0.001
	Region × Species	2	742.05	1.21	0.310
	Error	41			
Relative calc A	Region	2	0.51	29.56	<0.001
	Species	1	0.00	0.12	0.730
	Region × Species	2	0.01	0.75	0.477
	Error	41			
Relative conand	Region	2	1.02	42.66	<0.001
	Species	1	0.01	0.89	0.350
	Region × Species	2	0.01	0.40	0.673
	Error	41			

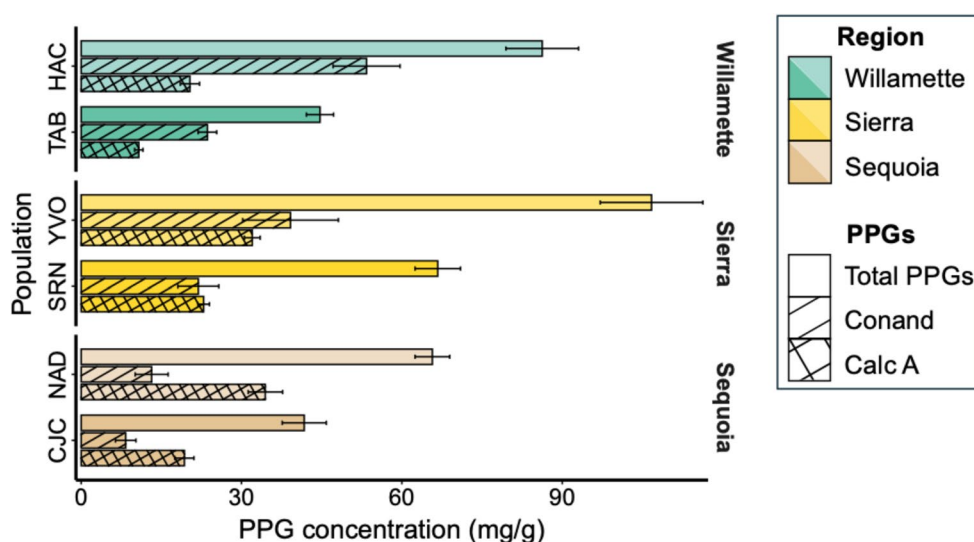
populations (Ontogeny × Population_{PPG arsenals}: pseudo $F_{2,42} = 0.23$; $P = 0.952$; Fig. 3). A closer inspection of arsenal variation across ontogeny revealed that PPGs only shifted along one of two NMDS axes (Ontogeny_{NMDS1}: $F_{1,42} = 13.11$; $P < 0.001$; Ontogeny_{NMDS2}: $F_{1,42} = 0.57$; $P = 0.453$; Table 1). For example, conandroside loaded almost entirely on NMDS1 and decreased in concentration in all populations later in development, whereas calceolarioside A loaded almost entirely on NMDS2 and maintained relatively constant concentrations in all populations across ontogeny

(Table S1 and S3). These data indicate that populations of *M. moschatus* consistently have higher concentrations of PPGs earlier in development and shift their chemical defense arsenals in similar ways as individuals mature, but that not all PPGs change in concentration across ontogeny.

Congeneric Comparison of Defenses

Geography and species identity influenced both total PPGs and relative proportions of the most prevalent PPGs. Consistent with previous results, total PPGs and the relative abundances of conandroside and calceolarioside A varied across latitude (Region_{Total PPGs}: $F_{2,41} = 13.78$, $P < 0.001$; Region_{Relative calc A}: $F_{2,41} = 29.56$, $P < 0.001$; Region_{Relative conand}: $F_{2,41} = 42.66$, $P < 0.001$; Table 2; Fig. 4). Across all populations, *M. guttatus* had consistently higher total PPGs (Species_{Total PPGs}: $F_{1,41} = 46.34$, $P < 0.001$; Fig. 4). However, relative concentrations of calceolarioside A (Species_{Relative calc A}: $F_{1,41} = 0.12$, $P = 0.730$) and conandroside (Species_{Relative conand}: $F_{1,41} = 0.89$, $P = 0.350$) were largely similar between *M. moschatus* and *M. guttatus*. Likewise, total PPGs (Region × Species_{Total PPGs}: $F_{2,41} = 1.21$, $P = 0.310$), relative calceolarioside A concentrations (Region × Species_{Relative calc A}: $F_{2,41} = 0.75$, $P = 0.477$), and relative conandroside concentrations (Region × Species_{Relative conand}: $F_{2,41} = 0.40$, $P = 0.673$) were consistent across species in every sampled region. Together, these results indicate that geographic patterns in chemical defense are largely conserved between species, despite species-level differences in total PPGs.

Fig. 4 Congeneric comparison of total PPGs and highest concentration PPGs in nearby populations. Colour represents the region of different population pairs. Lighter shades of similar colours represent *M. guttatus* populations from Kooyers et al. 2017 (HAC, YVO and NAD), whereas darker shades represent *M. moschatus* populations from the current study (TAB, SRN and CJC). Concentrations of total PPGs are represented by solid colours, conandroside (Conand) by striped hatching and calceolarioside A (Calc A) by cross-hatching. Error bars represent standard error



Discussion

Our results highlight ontogenetic and geographic variation in total concentrations and arsenals of chemical defenses in *M. moschatus*, as well as conserved relative concentrations and arsenals across space with its congener, *M. guttatus*. Specifically, our middle latitude population (SRN) had higher total PPGs compared to individuals from higher (TAB) and lower latitude (CJC) populations. Populations from different geographic origins also had distinct PPG arsenals. We observed that developmental stages strongly influenced total concentrations of PPGs, with greater investment in defense compounds in tissue produced early in life. The composition of PPGs also shifted ontogenetically, but did so similarly among populations. Notably, PPG arsenals of *M. moschatus* and its congener *M. guttatus* were locally similar and shifted in similar ways across space.

The observed patterns can be discussed in the context of established frameworks such as the ODT and LHDH. Support for the LHDH has been observed in *Asclepias* spp. and *Oenothera biennis*, with generally higher defense concentrations at lower latitudes (Rasmann and Agrawal 2011; Agrawal et al. 2012), sometimes despite higher levels of herbivory at higher latitudes (Anstett et al. 2014, 2015, 2016b). In our study, *M. moschatus* populations from lower latitudes had more individual compounds in their PPG arsenals. While this pattern fits the predictions of the LHDH, where higher herbivore pressure or more diverse communities drive greater diversity of compounds, we have not evaluated the selective pressures within these populations (Rotter et al. 2019). Spatial patterns in total concentrations of chemical defenses, on the other hand, did not follow the expectations of the LHDH, i.e., total PPGs were not highest in lowest latitude populations.

Our populations at the southern and northern extents of our sampling had lower concentrations of total PPGs (Figs. 1 and 2; Table S1). These phytochemical patterns match those found in *M. guttatus* (Kooyers et al. 2017; Rotter et al. 2022) and among *Mimulus* species (Holeski et al. 2021). However, we have not measured herbivore pressure in this study—a potential key factor that could drive our patterns (Pennings et al. 2009; Schemske et al. 2009). Likewise, we have sampled only three populations. Our results provide testable hypotheses for a study with greater population-level sampling across similar gradients in climate and herbivore pressure.

The ODT suggests that variation in total defense should coincide with development if shifts in phytochemical profiles increase individual fitness (McKey 1974; Rhoades 1979; Whittall 1999; Stamp 2003; but see Ochoa-López et al. 2020). Decreases in investment to chemical defenses later in development may be favoured in some species (Holeski

et al. 2012; Cole et al. 2021), as juvenile plants are generally more vulnerable to herbivory and thus selection may favour higher levels of defenses earlier in development. However, some species have higher investments later in development (Karban and Thaler 1999; Quintero and Bowers 2011, 2018; Blanchard and Bowers 2020) or more complex patterns of seasonal variation in defense (Carper et al. 2022). Our results suggest that investment in defense decreases during development in all populations, which is in contrast to a similar study of *M. guttatus* (Holeski et al. 2013). Differences in life-history strategies do not appear to explain these differences among species; however, inducible defenses or the type of herbivore pressure (i.e., specialist vs. generalist herbivores) may influence investment in chemical defenses throughout development. While these patterns suggest that *M. moschatus* constitutively adjusts defenses during development, more extensive research is needed into variation across tissue types and environments as well as assessments of associations between PPG concentrations, herbivory, and fitness.

Ontogenetic shifts in arsenals of *M. moschatus* varied among populations, with some consistent trends observed across populations. Of the major PPGs, conandroside consistently decreased while calceolarioside A stayed relatively constant through development. Most of the remaining PPGs decreased, however, a few did increase later in development. Results from similar studies in other species tend to be very species-specific. In *Datura* species as well as *Plantago lanciolota* and *Lonicera × bella* most if not all compounds increase in concentration throughout the growing season (Kariñho-Betancourt et al. 2015; Quintero and Bowers 2018; Blanchard and Bowers 2020). In oaks, defense arsenals do not appear to change ontogenetically (Moreira et al. 2020). However, our results resemble species where higher concentration compounds peak early in the season and gradually decline, while lower concentration compounds increase later (Carper et al. 2022). We note that exact methodologies differ between these studies and could influence the observed patterns. The developmental shifts we observed may be influenced by seasonal or herbivore-related factors, but these were not measured here. Our results highlight how defense traits are not static, but often vary in response to both internal developmental cues and external environmental factors (Barton and Boege 2017; Westerband et al. 2021).

On a broader evolutionary scale, closely related species are predicted to have more similar chemical defenses because of common descent and/or harboring similar herbivore communities (Ehrlich and Raven 1964; Marquis et al. 2016). Interestingly, total PPG concentrations were consistently lower in *M. moschatus* populations than in *M. guttatus* populations (Fig. 4). This was contrary to our expectations

since other *Mimulus* perennials tend to have higher levels of defense compared to their shorter-lived annual congeners, although there is limited phylogenetic signal for total PPG concentration across the *Mimulus* genus (Holeski et al. 2021). Differences in total PPGs may reflect alternative defense strategies, such as potential investment in physical defenses (e.g., trichomes), a hypothesis that should be tested in future studies.

However, there is a strong phylogenetic signal for PPG arsenals across the *Mimulus* genus—specifically, defense arsenals of more closely related *Mimulus* species tend to be more similar to one another (Holeski et al. 2021). This is consistent with our results where nearby pairs of elevation-matched populations of *M. moschatatus* and *M. guttatus* possessed relatively similar PPG arsenals with similar relative concentrations of key PPGs in both species from the same geographic region (Fig. 4). This pattern suggests that herbivore pressure and/or communities may be similar for both species in similar regions and diverge in parallel along the range. However, our results are inconsistent with patterns observed other species, such as the neotropical tree genus, *Inga*, where closely related species occurring in geographic proximity have evolved divergent defense arsenals (Kursar et al. 2009; Coley et al. 2018). This inconsistency may be due to *Inga* species experiencing herbivory from specialist herbivores within a relatively stable tropical climate whereas *Mimulus* species occupy a broader temperate distribution with high seasonality and experience herbivory from more generalist herbivores (Rotter et al. 2018, 2019). Together, these results highlight the need for further research into how chemical defenses vary among closely related species and populations, and the ecological or evolutionary factors influencing these patterns.

Conclusion

Our findings demonstrate there is substantial variation in chemical defenses of *M. moschatatus* across both the species' range and throughout development. While ontogenetic shifts in PPG concentrations are consistent with higher investment in defense at early developmental stages, these patterns should be interpreted cautiously given our limited sample size and lack of herbivory data. Geographic variation aligns more closely with environmental conditions than latitude-based herbivory predictions. The observed similarities in defense arsenals between *M. moschatatus* and *M. guttatus* suggest potential shared environmental influences, though differences in total concentrations suggest species-specific strategies exist. Together, these results provide a descriptive baseline and highlight the dynamic nature of plant chemical defenses, emphasizing the importance of

considering ecological context in interpreting patterns of phytochemical variation.

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Data Availability All code and data used throughout this manuscript are available on the GitHub page for SGI (https://github.com/innessim/musky_PPGs). Data and code have been permanently archived in DRYAD (<https://doi.org/10.5061/dryad.c866t1gkj>).

Declarations

Competing interests The authors declare no competing interests.

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